

The Utility Problem

By definition, the coincident peak (CP) is a customer's usage at the time of the system's maximum usage. In other words, CP is the demand for a particular selection of system users (e.g. customer(s), rate class(es)) at the time of the system's peak demand. Analysis requires the selection of a system user(s), a high-level comparator (usually the entire system or other segmentation criteria), and a time period. Each system user will have their own CP value, which equals their demand at the comparator's peak during the time period (comparator peak determines the time at which system users' demand value is taken). Non-coincident peak (NCP) is the customer's highest usage during the selected time period. NCP requires the selection of a system user(s) or group(s), and a time period. Each system user/group will have their own NCP value, which equals its peak demand during the time period.

Knowledge and insights into CP and NCP are essential for regulatory reporting; and overall awareness of demand contribution for different topological and geographical segments (feeder, substation, area, ect.) as well as different customer segments (rate, area, class, type, etc.). Insights originating from this type of analysis allow utilities to understand consumer behaviour and improve rate design to increase the satisfaction and retention of consumers. Lastly, with the rise of EVs and EV Fleets, CP and NCP are critical to understand to better size and manage assets.

The Data Solution and Method

Awesense Digital Energy Platform offers simplified ways to analyze the CP and NCP throughout the distribution area or customer segments. Awesense Platform's unique linkage of geospatial data (GIS) with time-series data (in this case, AMI or Billing data) enables this type of analysis. With data populated in the Awesense Platform, the Energy Data Model's (EDM) database views and functions will accelerate the building of CP and NCP analysis.

With Awesense EDM, the different methods for CP and NCP analysis are possible. These methods vary in the definition of system user groups. With EDM, the meter user groups for the analysis can be determined by multiple criteria:

1. Topological Criteria (using the connectivity tracing function calls):
 - a. Feeder level topology,
 - b. Substation level topology,
 - c. Transformer level topology,
 - d. Other GIS data topologies.
2. By hierarchical data from association attributes, like:
 - a. Substation level association,
 - b. MV Feeder level association,
 - c. MV/LV transformer association,
 - d. LV feeder association.

Coincident Peak and Non-Coincident Peak Analysis

[UC01-01] - Awesense Digital Energy Platform Use Case



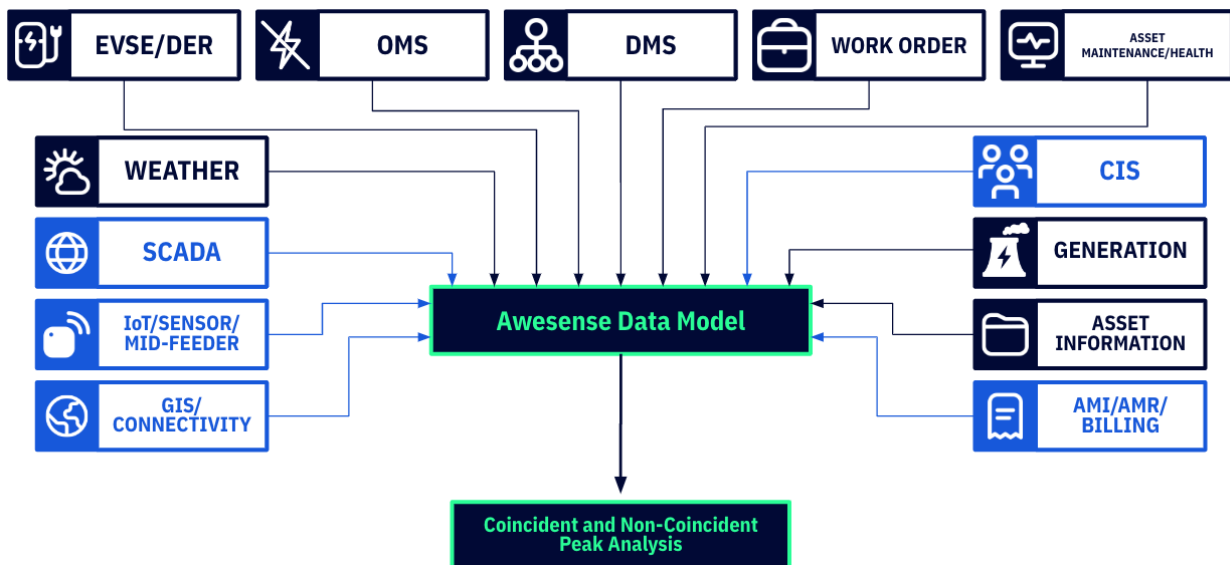
3. Geographical Criteria:
 - a. Zoning levels,
 - b. Geographical zoning (ZIP codes, counties, states, etc.),
 - c. Service area zoning.
4. Customer Segment Groupings:
 - a. By tariff group or rate class,
 - b. By customer type,
 - c. By technology class (with EV, with PV, etc).

Queries for stated groups can be created alone or combined.

On the time-series side, EDM allows for computing the aggregated time-series for defined groups or (if applicable) the parent measurement devices like SCADA data can also be pulled. This can be achieved by combining EDM's views for the `grid_element_data_source` and `ts_data_source_select` functions. From here, it is simple to determine the group's (system) peak and produce the required data for the entire group.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Data Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.